2. Models for Statistical Structured Prediction: Probabilistic Graphical Models

2.1. Directed Graphical Models (Bayesian Networks)

- Generative models: HMM and PCFG

2.2. Undirected Graphical Models (Markov Random Fields)

- Discriminative Models: Conditional Random Fields

Probabilistic Graphical Models

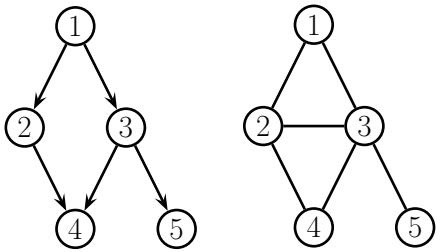
Probabilistic Graphical Models are an elegant framework that combines uncertainty (probabilities) and logical structure (independence constraints) to make simplifying assumptions about which variables affect which other variables.

These assumptions can often be represented graphically, leading to whole sets of models known collectively as Graphical Models (GMs)

Why do we need Graphical Models?

- GMs are the basis for the probabilistic approach to *Intelligent Systems*.
- GMs are a compact representation of joint probability distributions using graphs, constituting a perfect match between **Probability Theory** and **Graph Theory**.
- GMs allow us to abstract out the conditional independence relationships between the variables from the details of their parametric forms.
- GMs generalize to Neural Networks and Hidden Markov Models, among others.

Probabilistic Graphical Models



[from K.P. Murphy, 2012]

Nodes represent random variables.

The **graph** represents a set of independences and factorizes a distribution.

Graphical models: taxonomy

- Bayesian Networks based on **Directed Graphs**.
- Markov Random Fields based on **Undirected Graphs**.

Probabiistic Graphical Models

Key components

- **Inference**: deduce probability distributions from other given.
- **Learning**: obtain the probabilistic model from observations.

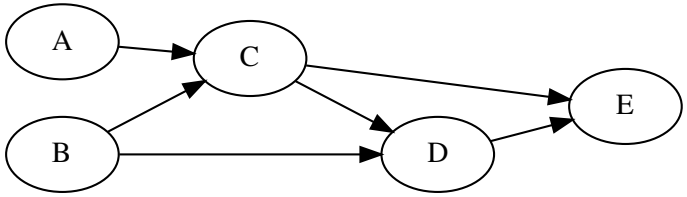
Applications

- Medical diagnosis, fault detection, ...
- **Computer Vision**: image segmentation, 3D reconstruction, scene analysis, ...
- **Natural Language Processing**: speech recognition, information extraction, machine translation, ...
- **Robotics**: planning, detection, ...

Bayesian Networks

A Bayesian Network (BN) is a graphical model that represents a set of random variables and their conditional dependencies via a Directed Acyclic Graph (DAG).

Example



Nodes represent to random variables (discrete or continuous)

Edges represent statistical dependencies between the variables

Bayesian Networks

- There are two extreme cases:

1) $$p(x_1, x_2, \ldots, x_T) = p(x_1) \cdot p(x_2|x_1) \ \ldots \ p(x_T|x_1, \ \ldots, \ x_{T-1})$$
$$= p(x_1) \prod_{t=2}^{T} p(x_t|x_1, \ldots, x_{t-1}) \qquad \text{requires } 2^T - 1 \text{ parameters}$$

2) $$p(x_1, x_2, \ldots, x_T) = \prod_{t=1}^{T} p(x_t) \qquad \text{requires } T \text{ parameters}$$

- If not all dependencies are possible, the complete factorization will be reflected in the graph (DAG) associated with the BN.

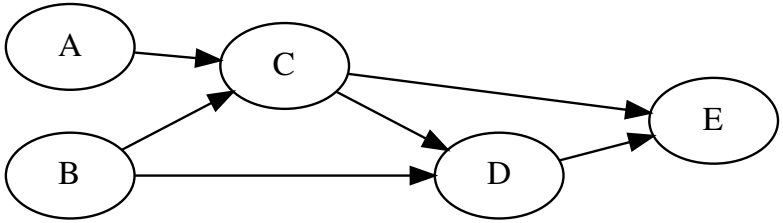
- A BN with nodes $x_1, \ldots, x_T$ defines a joint probability distribution:

$$p(x_1, x_2, \ldots, x_T) = \prod_{t=1}^{T} p(x_t \,|\, \varphi(x_t))$$

where $\varphi(x_t)$ denotes the dependencies associated with node x_t

Bayesian Networks

Example



$$p(A, B, C, D, E) = p(A) \cdot p(B) \cdot p(C|A, B) \cdot p(D|B, C) \cdot p(E|C, D)$$

BNs: a detailed example

$P(S \mid R)$

Rain (R)	Sprinkler (S) s	Sprinkler (S) r
n	0.60	0.40
y	0.99	0.01

$P(R)$

Rain (R) n	Rain (R) y
0.8	0.2

Sprinkler	s: stoped
	r: running
Grass	d: dry
	w: wet
Rain	n: not rain
	y: yes it rains

$P(G \mid S, R)$

Rain (R)	Sprinkler (S)	Grass (G) d	Grass (G) w
n	s	0.99	0.01
n	r	0.10	0.90
y	s	0.20	0.80
y	r	0.01	0.99

Joint distribution: $P(R, S, G) = P(R) \ P(S \mid R) \ P(G \mid R, S)$

Exercise: What is the probability that the sprinkler will work if there is no rain and the grass is wet?

BNs: conditional independence

Representing knowledge through the notion of conditional independence

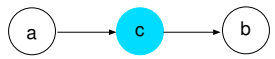
- Two events, a and b, are **(unconditionally) independent** and we denote it as $(a \perp\!\!\!\perp b)$ `if`:

$$p(a, b) = p(a) \cdot p(b) \qquad \text{or} \qquad p(a \mid b) = p(a)$$

- Two events, a and b, are **conditionally independent** given an event c, and we denote it as $(a \perp\!\!\!\perp b \mid c)$ `if`:

$$p(a \mid b, c) = p(a \mid c) \qquad \text{or} \qquad p(a, b \mid c) = p(a \mid c) \cdot p(b \mid c)$$

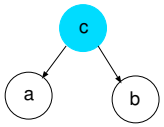
BNs: rules of conditional and unconditional independence



Causal direction: $(a \perp\!\!\!\perp b \mid c) \quad (a \not\perp\!\!\!\perp b)$

$$P(a, b \mid c) = \frac{P(a)P(c \mid a)P(b \mid c)}{P(c)} = P(a \mid c)P(b \mid c)$$

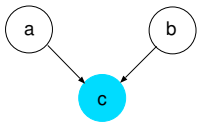
but $\quad P(a, b) \neq P(a)P(b)$



Common parent: $(a \perp\!\!\!\perp b \mid c) \quad (a \not\perp\!\!\!\perp b)$

$$P(a, b \mid c) = \frac{P(c)P(a \mid c)P(b \mid c)}{P(c)} = P(a \mid c)P(b \mid c)$$

but $\quad P(a, b) \neq P(a)P(b)$



V-structure: $(a \not\perp\!\!\!\perp b \mid c) \quad (a \perp\!\!\!\perp b)$

$$P(a, b \mid c) \neq P(a \mid c)P(b \mid c)$$

but $\quad P(a, b) = \sum_c P(a)P(b)P(c \mid a, b) = P(a)P(b)$

BNs: inference

The purpose is to estimate the posterior probability of some variable x from the joint distributions associated with a BN, given some evidence y (regardless of the values of the rest of the variables z).

$$P(x \mid y) = \frac{P(x, y)}{P(y)} \quad \text{con:} \quad P(x, y) = \sum_z P(x, y, z); \quad P(y) = \sum_{x, z} P(x, y, z);$$

The goal is to efficiently calculate $P(x, y)$ and $P(y)$.

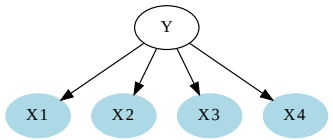
- The usefulness of calculating the posterior probability.

  Prediction.- What is the probability of observing a symptom knowing that the patient has a particular disease?

  Diagnosis.- What is the probability that a particular disease is a correct diagnosis given some symptoms?

BNs: examples (generative models)

Classifiers



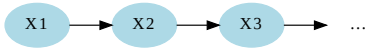
Naive Bayes

- Predicting a single class y given a vector of features $\mathbf{x} = \{x_1, x_2, \ldots, x_T\}$
- `Assumption`: once the class label is known all the features are independent

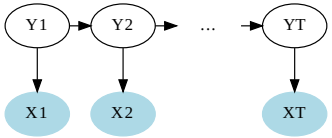
$$p(x_1, x_2, \ldots, x_T, y) = p(y) \prod_{t=1}^{T} p(x_t|y)$$

BNs: examples (generative models)

Sequential Prediction



Sequence Labeling



Markov model

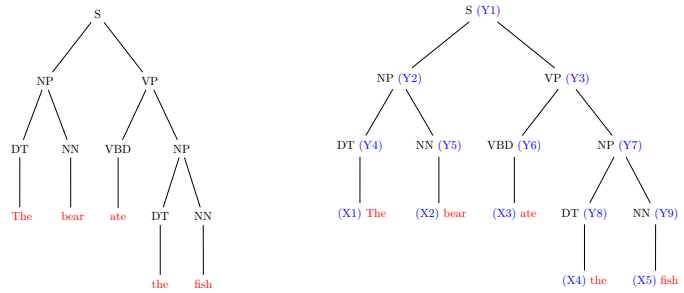
$$p(x_1, x_2, \ldots, x_T) \;=\; p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_2) \ldots \;=\; p(x_1) \prod_{t=1}^{T} p(x_t|x_{t-1})$$

(Bigram) Hiddden Markov Model

$$P(x_1^T, y_1^T) \;=\; \prod_{t=1}^{T} q(y_t|y_{t-1}) \cdot e(x_t|y_t)$$

Bayesian Networks: examples (generative models)

Parsing



Probabilistic Context-Free Grammars

$$t_x \;=\; [\, S = \alpha_0 \overset{r_1}{\Rightarrow} \alpha_1 \overset{r_2}{\Rightarrow} \alpha_2 \cdots \alpha_{m-1} \overset{r_m}{\Rightarrow} \alpha_m = \mathbf{x} \,] \qquad P(x_1^T, t_x) = \prod_{i=1\cdots m} P(r_i);$$

$$P(r_i) \;=\; P(Y_j\; Y_k|Y_i) \qquad \textbf{or} \qquad P(r_i) \;=\; P(X_l|Y_i)$$

Markov Random Fields

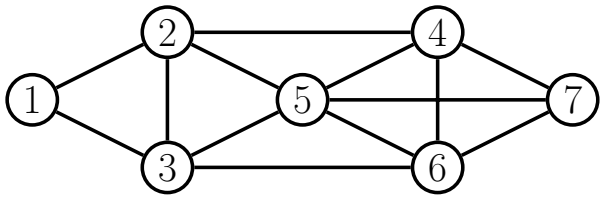
Bayesian Networks cannot perfectly represent all distributions.

Markov Random Fields (MRF) are graphical models based on Undirected Graphs.

The **nodes** represent the random variables.

The **edges** represent some notion of probabilistic interaction between neighboring nodes. Edges show which variables depend on each other.

Example [Kevin P. Murphy, 2012]



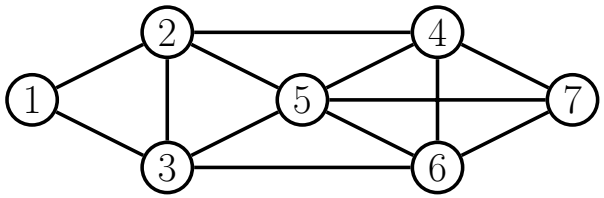
Markov Random Fields

Conditional independence properties: **global Markov property** for MRFs

MRFs define conditional independence relationships via simple graph separation: for sets of nodes A, B, and C, we say,

$$A \perp B \mid C \quad \Leftrightarrow \quad C \text{ separates } A \text{ from } B \text{ in the graph}.$$

Example:



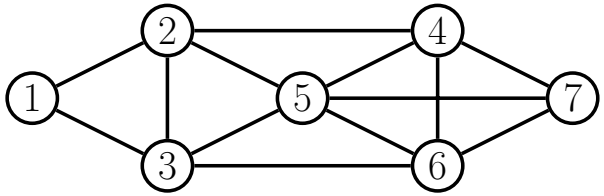
$$\{1,2\} \perp \{6,7\} \mid \{3,4,5\}$$

Markov Random Fields

Definitions: A **clique** of an undirected graph is a fully connected subgraph.

A **maximal clique** is a clique that is not a subgraph of any other clique.

Example:



Cliques: $\{1,2\}; \ldots \{1,2,3\}; \{2,3,5\}; \{2,4,5\}; \{3,5,6\}; \{4,5,6\} \ldots \{4,5,6,7\};$

No-cliques: $\{1,2,3,5\}; \{2,3,4,5,6\}; \ldots$

maximal cliques: $\{1,2,3\}; \{2,3,5\}; \{2,4,5\}; \{3,5,6\}; \{4,5,6,7\};$

Markov Random Fields: factorization

Definition.- Let G be the undirected graph that underlies an MRF. A probability distribution P_G defines a `factorization` over G if it is associated with:

$$P_G(X_1, \ldots, X_T) \;=\; \frac{1}{Z} \prod_{C \in Q} \psi_C(V_C)$$

Where,

$V = \{X_1, X_2, \ldots, X_T\}$ is the set of random variables;

Q is the set of all the (maximal) cliques of G;

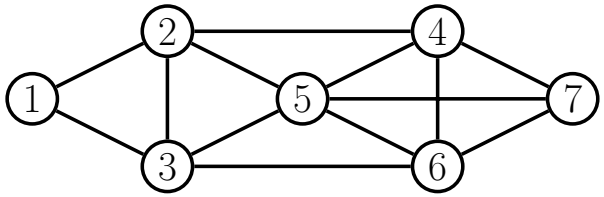
V_C is the subset of variables from clique C ;

$\psi_C : Q \rightarrow \mathbb{R}^{>0}$ is the **potential function** for clique c, and

Z is a normalization factor (constant) defined as:

$$Z \;=\; \sum_{X_1,\ldots,X_T} \prod_{C \in Q} \psi_C(V_C)$$

Markov Random Fields: factorization



Example $Q \;=\; \{\, \{1,2,3\};\ \{2,3,5\};\ \{2,4,5\};\ \{3,5,6\};\ \{4,5,6,7\};\ \}$

$$P_G(1,2,3,4,5,6,7) \;=\; \frac{1}{Z}\, \psi_1(1,2,3) \cdot \psi_2(2,3,5) \cdot \psi_3(2,4,5) \cdot \psi_4(3,5,6) \cdot \psi_5(4,5,6,7)$$

MRFs are more powerful than BNs but are more challenging to deal with computationally.

structured prediction: CRFs

Given $x \;=\; x_1, x_2, \ldots, x_T \;\in\; \mathcal{X}^*$ and $y \;=\; y_1, y_2, \ldots, y_T \;\in\; \mathcal{Y}^*$

- Discriminative models: **Conditional Random Fields (CRF)**

  `[Lafferty, McCallum, Pereira, 2001]`.

  - Model parameters θ
  - Compatibility function $\phi(x, y; \theta) \;\rightarrow\; \mathcal{R}$

    that that gives high positive scores to compatibles pairs (x, y)
  - Conditional distribution:

    $$p(y \,|\, x\,;\, \theta) \;=\; \frac{\exp\{\, \phi(x, y\,;\, \theta)\,\}}{Z(x\,;\, \theta)}$$

    Where

    $$Z(x\,;\, \theta) \;=\; \sum_{y' \in \mathcal{Y}^*} \exp\{\phi(x, y'\,;\, \theta)\}$$

CRFs: compatibility functions

Example [Carreras, 2011]

	1	2	3	4	5
x	Jack	London	went	to	Paris
y	PER	PER	-	-	LOC
y^1	PER	LOC	-	-	LOC
y^2	-	-	-	-	LOC

- (Bigram) Indicator Features

$$f_1(y_{t-1}, y_t, x_t) = \begin{cases} 1 & \text{if } x_t = \text{"London" and } y_{t-1} = \text{PER and } y_t = \text{PER} \\ 0 & \text{otherwise} \end{cases}$$

e.g., f_1 (PER, PER, x_2) $= 1$, and f_1 (PER, –, x_3) $= 0$

- Other possible examples of indicator features:

$f_1(\ldots) = 1$ **iff** $x_t =$"London" and $y_{t-1} =$ PER and $y_t =$ PER
$f_2(\ldots) = 1$ **iff** $x_t =$"London" and $y_{t-1} =$ PER and $y_t =$ LOC

$f_3(\ldots) = 1$ **iff** $x_{t-1} =\sim$/(in|to|at)/ and $x_t =\sim$/[A-Z]/ $y_t =$ LOC
$f_4(\ldots) = 1$ **iff** $y_t =$ LOC and $\text{WORLD_CITIES}(x_t) = 1$
$f_5(\ldots) = 1$ **iff** $y_t =$ PER and $\text{FIRST_NAMES}(x_t) = 1$

CRFs: factored compatibility functions

- Let $f(y_{i-1}, y_i, x_i)$ be a vector of features

$$(\, f_1(y_{i-1}, y_i, x_i),\ f_2(y_{i-1}, y_i, x_i),\ \ldots,\ f_K(y_{i-1}, y_i, x_i)\,)$$

- Given $f(.)$ and let $\theta \in \mathcal{R}^K$ be a weight matrix

$$\begin{aligned} \phi(x, y; \theta) &= \sum_{t=1}^{T} \theta\ f(y_{t-1}, y_t, x_t) \qquad (1) \\ &= \sum_{t=1}^{T} \sum_{k=1}^{K} \theta_k\ f_k(y_{t-1}, y_t, x_t) \end{aligned}$$

  - Let $y_0 = \text{NULL}$, so the bigram $y_0\, y_1$ is defined for $t = 1$.
  - This factorization will allow efficient algorithms since
    if $y \neq y'$ share bigrams then they will share scores

Conditional Random Fields

$$\begin{aligned} p(y|x;\theta) &= \frac{\exp\{\, \phi(x, y; \theta)\,\}}{Z(x;\theta)} \\ &= \frac{\exp\{\, \sum_{t=1}^{T} \sum_{k=1}^{K} \theta_k\ f_k(y_{t-1},\ y_t,\ x_t)\,\}}{Z(x;\theta)} \qquad (2) \end{aligned}$$

Where

$$Z(x;\theta) = \sum_{y' \in y^*} \exp\left\{ \sum_{t=1}^{T} \sum_{k=1}^{K} \theta_k\ f_k(y'_{t-1},\ y'_t,\ x_t) \right\} \qquad (3)$$

- Features $f(.)$ are given (problem-dependent).
- $\theta \in \mathcal{R}^K$ are the parameters of the model.
- CRFs are log-linear models on the feature functions.

CRFs: three problems

- Inference: Compute the probability of an output sequence y for x

$$p(y\,|\,x\,;\,\theta) = \frac{1}{Z(x\,;\,\theta)} \prod_{t=1}^{T} \exp\,\{\, \sum_{k=1}^{K} \theta_k\ f_k\ (y_{t-1},\ y_t,\ x_t)\,\}$$

- Decoding: predict the best output sequence for x

$$\arg\max_{y \in \mathcal{Y}^*} p(y\,|\,x\,;\,\theta) = \arg\max_{y} \prod_{t=1}^{T} \exp\,\{\, \sum_{k=1}^{K} \theta_k\ f_k(y_{t-1},\ y_t,\ x_t)\,\}$$

- Prameter estimation: learn parameters θ, given training data

$$\{\, (x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(N)}, y^{(N)})\,\}$$

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